NOT MEASUREMENT SENSITIVE

MIL-T-45380E(AT)
15 May 1991
SUPERSEDING
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29 May 1980

MILITARY SPECIFICATION

* TRUCKS, MAINTENANCE: EARTH AUGER, TURNTABLE BASE, DIESEL ENGINE DRIVEN, 28,000 POUNDS GVW, 4X4, MODIFIED COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers an earth auger with a turntable, mounted on a diesel engine driven, four wheel, four wheel drive, modified commercial truck chassis having a minimum gross vehicle weight (GVW) of 28,000 pounds for normal operating conditions. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to; US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	VV-F-800	- Fuel Oil, Diesel.

SPECIFICATIONS
MILITARY

*	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
*	MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
*	MIL-M-43719	- Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specification for.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS
FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

*	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, identification Marking and Data Plate Standards.
	MS51317	- Light, Warning, Vehicular, Rotating, D.C.
	MS51335	- Pintle Assembly, Towing – 18,000 Lbs. Capacity, Manual Release.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE (DOD)
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbing Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations
Federal Motor Vehicle Safety Standards.

* (Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) Standards:
* ASTM Designation: A53 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated Welded and Seamless, for Ordinary Uses (DoD adopted).

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices

*	J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers -Trucks, Truck-Tractors, and Trailers (DoD adopted).
*	J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
*	J537	- Storage Batteries.
*	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1,000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
*	J588	- Turn Signal Lamps.
*	J589	- Turn Signal Switch.
*	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
*	J683	- Tire Chain Clearance -Trucks, Buses, and Combinations of Vehicles.
*	J688	- Truck Ability Prediction Procedure.
*	J704	- Openings for Six- and Eight-Bolt Truck Transmission Mounted Power Take-Offs.
*	J706	- Rating of Winches.
*	J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
*	J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
*	J1349	- Engine Power Test Code -Spark Ignition and Diesel.

* (Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION. INC
Year Book

* (Application for copies of Tire and Rim Association publications should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

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* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informationa1services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehic1e and accessories. Except as specified in 3.1.1 through 3.1.1.13, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.13, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering office of the procuring activity prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and component specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223.

* 3.1.1.3 Rustproofing. When specified (see 6.2) the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

* 3.1.1.4 Drain plugs. Drain plugs installed in manual transmissions, transfer case and rear axle shall be of the permanent magnet type.

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* 3.1.1.5 Towing devices. Towing devices consisting of two hooks, loops, eyes or pins or the chassis manufacturer's standard single center mounted eye or pin shall be mounted on the front of the vehicle. When specified (see 6.2), in addition, towing devices shall be mounted on the rear of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

* 3.1.1.6 Trailer towing package. When specified (see 6.2), a trailer towing package shall be furnished. The trailer towing package shall consist of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring, and shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS51335-1. The pintle shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 1,800 pounds and a horizontal drawbar load of not less than 16,000 pounds directly to the chassis rails. The rearmost portion of the pintle shall be forward, but no more than four inches forward, of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The attachment devices shall be capable of accommodating a standard grab hook (2-3/16 inches wide, 47/64 inch thick and 31/64-inch throat width) for a 3/6-inch chain. The lighting receptacle conforming to SAE J560 with its conductors connected and color-coded as specified herein, or number coded, shall be mounted in a readily accessible location near the pintle.

* 3.1.1.6.1 Pintle height. When specified (see 6.2), the pintle height, measured to the centerline of the pintle, shall be 20 inches (plus 5 minus 0). The pintle shall be installed with bracket reinforcements to transfer pintle loads directly to the chassis rails. The installation shall be capable of withstanding a towed load equal to the GVW of the truck without permanent deformation.

* 3.1.1.7 Wheels splash and stone throw protection. The vehicle shall have rigid splash shields ahead of the rear wheels and rubber mud flaps to the rear of the rear wheels. All tilt cabs shall have rubber mud flaps to the rear of the front wheels. A metal strip not less than 0.125 inch thick and not less than 1 inch wide, extending the entire width of the mud flap shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.250 inch wide by 0.094 inch thick shall be furnished at each bolt. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

* 3.1.1.8 Trailer lighting cable. When specified (see 6.2), a trailer lighting cable conforming to SAE J1067 shall be furnished. The cable shall be coiled and shall be not less than 110 inches long when fully extended. Both ends of the cable shall be equipped with a round

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plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the vehicle tool compartment.

* 3.1.1.9 Spark arrester. As specified herein (see 3.4.4.1), the vehicle shall be furnished with an exhaust system spark arrester.

* 3.1.1.10 Brushguard. The vehicle shall be equipped with a radiator and head lamp brushguard. When the headlamps are recessed into and protected by the front bumper, a headlamp brushguard is not required.

* 3.1.1.11 Decals and data plates. The power takeoff caution notice shall conform to type I or type II, class 1 of MIL-M-43719 or composition A (class l or 2) or composition C of MIL-P-514.

* 3.1.1.12 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

* 3.1.1.13 4x4 conversion. The chassis manufacturer's standard 4x2 truck chassis may be modified to provide all wheel drive conforming to the requirements specified herein if:

(a) The conversion axle manufacturer's engineering department specifically approves and certifies that all such modifications meet the design requirements and standards of the conversion axle manufacturer. Certification shall be based on both design analysis and proving ground test reports, which shall be made available to the engineering and quality assurance offices of the procuring activity.
(b) The chassis manufacturer's front axle before conversion has the same load rating as the conversion axle to be installed.
(c) Components used in the all wheel drive conversion are of current production.
(d) Components used in the all wheel drive conversion are approved for the conversion application by the component manufacturers.
(e) The converted vehicle is certified to conform to Federal Motor Vehicle Safety Standard No. 121, by the intermediate or final manufacturer.

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(f) Replacement headlights, if required to be added, shall meet the height requirement of not less than 22 inches and not more than 54 inches, measured above the road surface, in conformance with Federal Motor Vehicle Safety Standard No. 108. Replacement headlights shall be equivalent in mounting, protection, and range and precision of adjustment to the chassis manufacturer's original standard headlights.

(g) Unused headlight cavities are covered in a neat workmanlike manner treated and painted to match the chassis cab color with treatment and painting equivalent to the chassis cab manufacturer's process for the remainder of the chassis cab. Cavities and their covers shall be rustproofed in accordance with 3.1.1.3.

(h) Complete installation drawings are available to the procuring activity.

3.2 General design.

3.2.1 Federal Motor vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with the Environmental Protection Agency regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicles Engines in effect on the data of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in affect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound 1evel shall not exceed 84db(A) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to Environmental Action Agency Noise Emission Standards for transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; body; and full complement of fuel, lubricants, and coolant.

3.2.5 Gross vehicle weight. The GVW shall consist of curb weight operator weight (computed at 175 pounds), and a payload to provide not less than the specified GVW. A vehicles with crew (four-door) cab shall have the GVW computed with 525 pounds of payload in the rear seat.

3.2.6 Weight distribution. The distribution of the GVW for the purpose of establishing suspension, axle, and tire capacities shall be determined with the payload uniformly distributed over the load area.

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* 3.2.7 Dimension. The cab-to-axle dimension shall be not less than 102 inches but may be reduced not more than 2 inches when the vehicle is furnished with a tilt type cab.

* 3.2.8 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity. The minimum GVW rating shall be 28,000 pounds.

* 3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.10 Asbestos. Asbestos materials shall not be used in any form in any part of the vehicle.

3.3 Performance. Performance requirements shall be met with the front, wheel drive disengaged and with front wheel driving hubs unlocked, except that vehicles equipped with interaxle compensating devices shall meet performance requirements with the front wheel drive engaged.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with the truck loaded to specified GVW.

3.3.1.1 High speed gradeabiliy. The vehicle shall ascend a continuous grade of 1.0 percent at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, and shall provide not more than the vehicle speed (mph) specified in table I for the corresponding number of available forward speeds.

TABLE I. Low speed requirements.

Number of forward speeds	Speed (mph)
8	3.0
9 or 10	2.5

3.3.1.3 Maximum geared speed. The maximum geared speed at the engine governed speed shall be not less than 52 mph (minus 1 percent). Conformance to the geared speed specified shall be determined by calculating in accordance with the following formula:

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$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than four cylinders. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-B fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-B is acceptable.

* 3.4.1.2 Oil filter. A full flow or combination full flow and bypass oil filter with replacement element shall be furnished.

3.4.1.3 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the maximum recommended operating speed.

* 3.4.1.4 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (^{o}F). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity.

* 3.4.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles the control shall include complete thermostatic control of all coolant flow through the radiator.

* 3.4.1.6 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

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* 3.4.1.7 Silicone rubber hoses. When specified (see 6.2) silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.8 Powerplant heaters and fuel warmers. When specified (see 6.2) a coolant heater, an engine oil heater and a fuel warmer (diesel engine driven vehicles only) shall be provided. Unless otherwise specified (see 6.2), a battery heater shall be provided when power plant heaters are specified. Heaters shall operate on 110 volts alternating current (ac) and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male) weatherproof, slave receptacle for receiving external power and grounding the vehicle. A three-wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle end and a standard three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to minus 60°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

* (a) A coolant heater, 1,000-watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. An engine thermostat with an operating range of 170°F to 195°F shall be installed.
* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:
  (1) Not less than 22 watts per quart heating capacity
  (2) Not more than 18 watts per square inch heating capacity
  (3) Thermal balance design or thermostat control providing for uninterrupted operation
  (4) Provision for mounting below minimum service oil level.

  (c) The battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperature as low as minus 60°F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

  (d) A fuel warmer or pre-heater shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of minus 40°F to an outlet temperature of plus 9°F, with a fuel flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

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* 3.4.1.9 Fuel fired engine preheater. When specified (see 6.2), for diesel engine driven vehicles, a diesel fuel fired engine water heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat and a circulating pump and shall be connected to the engine coolant system. The heater shall be capable of starting and operating at minus 60°F and shall heat the engine to plus 40°F from minus 60°F in not more than 1 hour. The system shall be equipped with a light, visible to the driver, to indicate that the preheater is operating.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. A 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or electric grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished.

* 3.4.2.2 Alternator. Unless otherwise specified (see 6.2) on diesel engine driven vehicles, a minimum 75-ampere alternator shall be furnished. The alternator output with the engine at engine idle speed shall be not less than 45 amperes.

* 3.4.2.3 Lighting. All vehicle lights, reflectors, and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.13, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

* 3.4.2.4 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall be self-cancelling and shall have a visible and audible type flash indicator.

* 3.4.2.5 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity ratings and the total cold cranking ratings at 0°F, both measured in accordance with SAE J537, shall be not less than specified in table II. The batteries shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

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TABLE II. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	320	1,200

* 3.4.2.6 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) shall be of not less than 50 gallons total capacity (43 gallons for vehicles destined for California). When more than one tank is furnished, means shall be provided to assure an equalized fuel level in both tanks.

* 3.4.3.3 Fuel and water separator. The manufacturer's standard or optional fuel filter shall be provided. When specified (see 6.2), a fuel and water separator shall be furnished. The separator shall include a water coalescer and a drain valve. When a separator is required, a combination filter/separator unit may be furnished.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. When vertical exhaust mufflers are furnished, a heat shield and a hinged rain cap shall be provided.

* 3.4.4.1 Spark arrester. When specified (see 6.2), a spark arrester shall be furnished, except on vehicles with turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

3.4.5 Gear train.

* 3.4.5.1 Transmission. Unless otherwise specified (see 3.4.5.3) a manually shifted transmission shall be provided on the vehicle. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall be provided with power takeoff opening(s) conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds. The transmission and axle gear ratios shall be selected to provide the performance specified in 3.3.1 through 3.3.1.3.

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3.4.5.1.1 Speeds. The vehicle shall be provided with not less than eight forward speeds. The vehicle may be furnished with an auxiliary transmission.

* 3.4.5.1.2 Transfer case. A transfer case shall be furnished. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between front and rear axles, the transfer case shall provide for driver selection of either two-wheel or four-wheel drive. When furnished, interaxle compensating devices shall provide for positive transfer of power to all driving axles. When a two-speed transfer case is furnished the speedometer shall read accurate speed with the speed selector in high or low range.

* 3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the type and class vehicle and engine furnished with torque capacity exceeding the maximum delivered engine torque. The clutch lining shall be asbestos free.

* 3.4.5.3 Automatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic transmission. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with a power takeoff opening. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction, in proportion to the number of driving axles furnished.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the vehicle furnished. Chassis frame rails shall not project beyond the rear end of the body.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member measured at the ground with the vehicle loaded to specified GVW. When the suspension capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic double acting shock absorbers at the front wheels.

* 3.4.9 Axle. Axle ratings shall be at least equal to the load imposed on each measured at the ground, with the vehicle loaded to specified GVW. When specified (see 6.2), the wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

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* 3.4.9.1 Traction control. When specified (see 6.2), the vehicle shall be furnished with traction control on the rear axle. The traction control shall actuate automatically to ensure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires and tubes. Unless wide base tires are specified, the vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in procurement documents, multi-piece rims shall not be furnished. Tire type and size tire load range (ply rating), and rim sizes shall be the same for all wheels on each vehicle, except for trucks furnished with a snowplow or snowplow provisions. When specified (see 6.2), wide base type tires and wheels for the front and rear axles shall be provided in lieu of conventional front and dual rear wheels and tires. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), 10-stud disc type wheels shall be furnished.

* 3.4.10.1 Tires. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Unless otherwise specified, standard profile tires shall be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall have nondirectional mud and snow tread. Unless otherwise specified (see 6.2), tires shall be of the tubeless type. Tires shall be of rated capacity at least equal to the load-imposed on each tire, measured at each wheel at the ground, with vehicles loaded to specified GVW. Tires and tire size designation systems shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.4.10.2 Inner tubes. When tube type tires are specified in acquisition documents, inner tubes shall be of the heavy duty type, and shall be of the proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.10.3 Carrier for spare tire assemble. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of, or plated with, corrosion-resistant material. The carrier design shall enable safe removal or mounting of a spare wheel assembly using only the tools specified in 3.4.16.1. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

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* 3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare wheel or rim or a spare tire assembly is furnished on a vehicle without a carrier, it shall be stowed securely on the vehicle for shipment. The spare wheel or rim shall be of the same size as that furnished on the front axle of the vehicle. When a spare tire assembly is specified a spare wheel or rim shall be furnished.

* 3.4.10.5 Spare tire assembly. When specified (see 6.2), an inflated spare tire, mounted on the spare wheel or rim shall be furnished. The spare tire shall be of the same size, type and tread design and load range (ply rating) as the tires furnished on the front axle of the vehicle.

* 3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

* 3.4.11.1 Air brakes. The vehicle shall be equipped with a four-wheel brake system consisting of full-air-application brakes. The brake system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 7.250 cubic feet per minute (cfm)
(b) Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
(i) Automatic slack adjusters on cam type brakes or internal self adjusting brakes on wedge and disc type brakes.

* 3.4.11.2.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not leas than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

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* 3.4.11.3 Trailer brake control system. In addition to components specified in 3.4.11.2 and 3.4.11.2.1, when specified (see 6.2), a trailer brake control system shall be furnished. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Prime mover protection valve with dash control and automatic breakaway feature (with air brakes)
(e) Trailer stoplight control operative with foot brake and with hand control for trailer brakes
(f) Two SAE J844 coiled air hoses, not less than 110 inches long, with SAE J318 gladhand couplers on both ends of the hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment
(g) Air connectors for trailer with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to prevent interference with the trailer
(h) Dummy gladhand couplers with chains
(i) Prime mover only parking brake valve to permit truck parking brakes to be applied while charging the trailer air brake system (with air brakes).

* 3.4.11.4 Brake control for use from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, a relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), and additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism; or a vehicle with a butterfly hood and with bolt on fenders capable of being removed or swinging forward when Unbolted; shall be furnished. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with external, key-operated lock. Full length drip rails shall be installed above the cab doors.

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* 3.4.12.1 Cab interior. Unless otherwise specified, the cab shall have an upholstered, full-width, adjustable seat and back or individual, adjustable driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable driver's seat and an individual passenger seat shall be provided. When specified for vehicles with air brakes (see 6.2), the driver's seat shall be the manufacturer's standard air ride suspension type. Interior lighting shall be provided. Seat belts shall be installed.

* 3.4.12.2 Crew cab. When specified (see 6.2), a four-door full width crew cab shall be furnished in lieu of the standard full width cab. The cab shall be equipped with two upholstered full width seats and backs. The front seat shall be adjustable. Three pairs of seat belts shall be installed for both the front and rear seats. The cab doors shall be equipped with locks operable from inside the cab through mechanical linkage, with both front doors equipped with external key-operated locks. Cab doors shall have windows with crank-operated window regulators. Interior lighting shall be provided. The cab roof shall be of one piece construction; or, if welded, the roof shall give the appearance of one piece, with weld seams being continuous, waterproof, and free of visible seams, bumps or protrusions. Full length drip moldings shall be mounted above the doors. When a crew cab is furnished, the cab-to-axle dimension shall be measured from the back face of the crew cab.

3.4.12.2.1 Seat spacing. With the front seat adjusted to the extreme forward position, there shall be not less than 30 inches, measured in a horizontal plane, between the front of the rear seat back and the rear of the front seat back. A kick space height of not less than 2.750 inches shall be maintained between the floor and the rear of the front seat in all positions of adjustment. Leg room and space forward of the front seat shall be equivalent to that provided ahead of the seat in a two-door standard cab.

3.4.13 Steering. Power steering shall be furnished.

3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by air or electric motor(s).

3.4.15 Bumper. Unless the bumper is an integral part of the vehicle cab, a channel-type front bumper shall be provided, except when a front mounted winch is furnished (see 3.4.22.2).

3.4.15.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86.

* 3.4.16 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains, and emergency reflective triangles shall be furnished. The stowage space shall provide for positive retainment of equipment during vehicle operation. Stowage

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space of these tools may be furnished inside the cab. When stowage space for the tools is located outside of the cab, it shall be waterproof and shall provide for locking with a padlock.

3.4.16.1 Too1s. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly and shall include at least a jack, jack handle, and wheel nut wrench. The jack shall be of such closed height as to permit its location under axle or other satisfactory lift point at any wheel with the tire flat. The jack without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of a wheel and tire assembly.

3.4.17 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted and visible to the driver when seated in driving position.

* 3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or red indicator warning light
(e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
(f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Ash receptacle
(i) Dual sunvisors
(j) Front door mounted armrest on driver’s side on model 4x2 and 4x4
(k) Driver's compartment ventilator other than window
(l) Tachometer (for diesel engine driven vehicles).

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When specified (see 6.2), an ammeter or voltmeter gage, oil pressure gage, and coolant temperature gage shall be provided in lieu of the indicators and buzzers specified above. When specified (see 6.2), an automatic engine shutdown system also shall be furnished. The system shall include an engine coolant temperature, engine coolant level and engine oil pressure red indicator warning light and alarm buzzer. The warning light and buzzer actuation shall preceed engine shutdown. The system shall permit engine 'restart and run for approximately 30 seconds following automatic shutdown. Also, engine oil pressure and engine coolant temperature gages shall be provided with the automatic engine shutdown system.

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall have flat and convex areas. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 25 square inches of reflective area and a radius of curvature of not less than 20 inches. The mirrors shall have not less than two supporting arms.

3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition an air operated horn shall be furnished.

* 3.4.22 Front mounted winch. When specified (see 6.2), the vehicle shall be furnished with a winch assembly, mounted on the front of the vehicle and powered by the auxiliary unit with controls at operator station and in the cab. The winch shall conform to SAE J706. The winch shall be single drum type and shall have one forward and one reverse speed. All winch controls shall be mounted inside the cab. Winch controls shall be located to provide no interference with the entrance or exit of the driver. The winch shall have a rated single line pull capacity of not less than 20,000 pounds on the bare drum. The winch line speed on the bare drum shall be between 15 and 20 feet per minute at an auxiliary engine speed equal to 1,500 RPM. The winch shall be wound with not less than 200 feet of 5/8-inch diameter, preformed 6x37, improved plow steel, independent wire rope core (IWRC) cable and shall be equipped with an end chain and hook. An integral, adjustable automatic safety brake shall be provided. The winch shall be equipped with a roller guide. Angles of the winch driveline U-joints shall be not greater than 16-1/2 degrees.

3.4.22.1 Winch drum guard. A winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges. The guard shall consist of not less than 0.250 inch vertical side plates conforming to the outside radius of the drum flanges. Six bars, 0.375 inch by 1.2450 inches shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius and three bare shall be located on the bottom and spaced equally on the bottom radius. The horizontal distance between the vertical side plates and the drum flanges shall be not greater than one half the specified cable diameter.

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* 3.4.22.2 Front bumper. When a front mounted winch is specified (see 3.4.22), the bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be either a channel (see 3.4.15), or a pipe type front bumper. When a pipe type front bumper is furnished the nominal diameter shall be not less than 3 inches, and the wall thickness shall be not less than that specified in Schedule 40 of ASTM A53. The pipe type front bumper shall have half-round ball ends.

3.4.22.3 Combination step plate and gravel guard. When a front mounted winch is specified a combination step plate and gravel guard shall be furnished. The step plate shall be fabricated of not less than 14 gage (0.0747 inch) steel tread plate exclusive of projections. The step plate shall be secured to the front bumper and shall be not less than 0.250 inch and not greater than 0.375 inch from the chassis sheet metal. The step plate shall be capable of supporting 300 pounds per square foot. The step plate shall not deflect more than 0.125 inch under the loads imposed.

* 3.4.23 Engine hour meter. When specified (see 6.2), an engine hour meter shall be furnished. The meter shall have a totalizing mechanism of not less than 9,999 hours for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to ensure continuous trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

* 3.4.24 Back-up alarm. When specified (see 6.2), the vehicle shall be provided with an audible, pulsating, signaling device (electric or mechanical) to caution personnel when the vehicle is in reverse gear operation.

* 3.4.25 AM/FM radio. When specified (see 6.2), the manufacturer's standard AM/FM radio shall be provided.

* 3.5 Earth auger unit. The earth auger unit shall be self-contained, turntable mounted unit, consisting of an auxiliary engine, hydrostatic drive system, boring assembly, pole setting equipment, augers, and self-contained winch mounted on the curbside of the truck. The main frame shall include a right angle drive trunnion support for extra strength and extend the life of right angle drive/final drive.

* 3.5.1 Auxiliary engine. An auxiliary engine to drive the hydraulic pump shall be furnished. The auxiliary engine shall operate on the same type of fuel as the truck chassis engine. The auxiliary engine fuel supply shall be from the auxiliary engine fuel tank, located streetside on the turntable. The engine shall be mounted on the body platform in conjunction with the hydraulic pump. The engine mounting shall minimize the transfer of vibration. The engine shall be capable of being operated from a remote position (see 3.5.7). The auxiliary engine shall have sufficient output power to drive all operating components other than the

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vehicle. During normal operating conditions, the continuous-duty horsepower output required by all components and accessories driven by the engine shall not exceed 75 percent of the full throttle rating of the engine. The size and weight of the engine assembly shall be held to a minimum consistent with the performance requirements specified herein.

* 3.5.1.1 Engine components. The engine shall be a four cylinder diesel engine. The engine shall be equipped with at least the following components:

(a) Self-starter, 12-volt
(b) Air cleaner
(c) Low oil pressure high water temperature, shutdown system
(d) Fuel water separator system.
(e) An engine hour meter, having a totalizing mechanism of not less than 9,999 hours
(f) When power plant heaters are specified (see 3.4.1.6) a 300-watt immersion type heater with 170°F to 195°F thermostat shall be installed in the oil pan through any convenient opening.

* 3.5.1.2 Engine control panel. An engine accelerator, a key-operated starting (ignition) switch, and a palm button starter shall be installed in a weather- and waterproof cabinet on a control panel, and each shall be identified as to its function. The cabinet shall be provided with a hinged cover with a latch and provisions for locking with a 1-inch padlock.

* 3.5.1.3 Fuel pump. A fuel pump shall be furnished on the engine, if required. When lifting fuel against a head of 4 feet, the pump shall have sufficient capacity to supply the engine at its maximum rated load. An in-line fuel filter shall be provided.

3.5.2 Hydrostatic drive system. A hydrostatic drive system shall be furnished to provide infinitely variable speeds without the use of an intermediate gearbox or shifting of gears throughout the speed range. The system shall develop a torque from 0 to 10,000 pounds. The system shall have pressure override protection built into the pump for kelly bar and ring and pinion gear sets. The system shall provide auger revolution range 0 to 180 revolution per minute (rpm). The system shall use one common reservoir and hydraulic fluid.

3.5.3 Boring assemble.

* 3.5.3.1 Performance. The auger shall bore holes up to 36 inches in diameter and 10 feet in depth in permafrost soil. The auger shall bore holes with a minimum variance of 53 degrees from extreme rightmost to extreme left most positions with either a minimum of 30 degrees to the right from center and 23 degrees to the left, or with a minimum of 23 degrees to the left from center and 30 degrees to the right. The auger shall bore holes up to a minimum of 20 degrees

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from the vertical off the rear of the truck. The auger shall bore holes up to 10 degrees from the vertical toward the truck and handle poles up to 75 feet in length. Depth of holes shall be measured from the ground level to the top of the loose dirt at the bottom of the hole.

3.5.3.2 Leveling mechanisms. The boring assembly shall be equipped with a hydraulic leveling device to permit the setting of the auger in a vertical position regardless of the tilt of the truck. Leveling devices shall permit elevating the auger shaft to the operating position and lowering to the traveling position.

* 3.5.3.3 Stabilizing supports. Two hydraulically operated modified "A" frame stabilizing supports shall be furnished to support the truck during drilling operations. The supports shall have a minimum spread of 125 inches.

* 3.5.3.4 Auger. The auger shall be raised, lowered, rotated, and forced into the earth by a hydraulically controlled mechanism. A minimum 6,500 pound capacity, hydraulically operated, telescoping derrick, sheave, and pole guide shall be furnished. A trombone tube assembly shall be provided to supply hydraulic oil to the pole guide and prevent hose damage. Exposed hoses not protected/trombone enclosed are not acceptable.

* 3.5.3.5 Auger bits. One each, Pengo heavy duty, twin helix type auger bit or equal, of the following sizes shall be furnished: 16-inch, 24-inch, 30-inch, and 36-inch. Bullet ripper teeth, shank plates and pilot bit shall be included.

3.5.4 High speed throw-off. A high speed throw-off shall be provided. Operation of the high speed throw-off shall not be dependent on shifting the transmission. The throw-off speed shall be sufficient to clear soil from the auger bit and deposit it so that when the auger is digging maximum depth holes, the soil will not fall back into the hole.

* 3.5.5 Auger unit winch. A heavy duty, single drum winch having the rated single line pull capacity of not less than 10,000 pounds on the bare drum shall be furnished. The winch shall have not less than one forward and one reverse speed. The winch shall conform to SAE J706. An integral, adjustable, automotive safety brake shall be provided. The winch shall be wound with 150 feet of 7/16-inch, preformed, 6x37, improved plow steel, wire rope having nominal breaking strength in excess of 10,000 pounds. The winch shall be direct coupled powered (winches supplied with chains and sprockets not acceptable) by the auger engine to permit independent operation of the auger or winch. A winch line hook shall be provided.

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* 3.5.6 Turntable. A turntable base shall rotate the unit through not less than a 180-degree arc and shall extend or retract the unit at a distance of not less than 28 inches from any point within the arc.

3.5.7 Operator's seat. An operator's seat shall be provided. The seat shall be located to permit control of and to provide a clear view, for the operator, of all boring operations. The seat shall be located convenient to all controls.

3.5.8 Warning light. An amber rotating, vehicular, warning light conforming to MS51317-4 shall be provided. The rotating light shall be mounted with reinforcement at the point of attachment on the vehicle centerline for 360-degree visibility. The warning light shall be operable by a separate switch with a red indicator light located on the instrument panel and marked to indicate the function of the switch.

* 3.5.9 Storage box. A material storage box 18 inches x 18 inches x 18 inches shall be supplied curbside. The box shall include drop down door, slam action lock with key, and weather stripping.

* 3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle for operational use including at least the following: alignment of lights; adjustment of the engine and electrical and brake systems; filling and charging of the battery; alignment of front wheels; inflation of all tires; complete lubrication of the chassis, engine, and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of the windshield washer reservoir with water and appropriate additives.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of industry.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of the product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on front and rear axles will be computed by the contractor and verified by the Government using the curb weight, the operator weight at 175 pounds and the payload (see 3.2.6) to provide the specified GVW. Calculated imposed loads on front and rear axles shall be compared to the suspension, axle, and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

* 4.3.2 Road test. The vehicle, for models built to 4 wheel drive while on the chassis manufacturer’s production line shall be road tested by the contractor without payload. The road test shall be for not less than 10 miles at speeds up to 52 mph. The vehicle, for 4 wheel drive models qualifying under 3.1.1.13, shall be road tested with and without payload. Payload shall be distributed first, so that the front axle is loaded to GAWR and the vehicle is loaded to GVW and second, with the rear axle loaded to GAWR and the vehicle loaded to GVW. The road test for each of the three conditions shall be for not less than 30 miles at speeds up to 50 mph over highways and gravel roads and for not less than 5 miles at speeds up to 15 mph over cross country terrain with ground and grade requiring all wheel drive. During the loaded and empty

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portions of the road test, the brakes shall be applied firmly, bringing the vehicle to a sudden stop not less than 5 times during each portion of the road test. During the road test, the front and rear suspension and the drivetrain shall be periodically inspected for interference and contact with other vehicle components. Abnormal contact of the drivetrain or suspension components with other components shall be cause for rejection. Front spring bumpers shall not make contact with frame stops except under the most extreme cross country conditions. Operational or mechanical failures of vehicle components during the road test shall be cause for rejection. Failure includes permanent deformation as well as breakage.

* 4.3.3 Truck body treatment and painting. A certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to the Government representatives at the first production vehicle inspection.

4.3.4 Boring assembly test. The boring assembly shall be operated to demonstrate compliance with 3.5.3.1 through 3.5.7. Tests may be performed in soils found in the vicinity of the contractor's plant.

4.3.5 Production sample. Upon acceptance of the first production vehicle the vehicle shall remain at the manufacturing facility as a production sample and shall be the last vehicle shipped on the contract contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

* 4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements specified herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment from the manufacturer's plant to the initial receiving activity in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended for general non-tactical use by the Government in the performance of the maintenance and construction tasks indicated.

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* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Identification of appropriate military service for painting (see 3.1.1.1).
(d) Identification of appropriate military service for marking (see 3.1.1.2).
(e) Concealed markings, if required (see 3.1.1.2).
(f) Rustproofing, if required (see 3.1.1.3).
(g) Tropical rustproofing, if required (see 3.1.1.3).
(h) Towing devices if required on rear (see 3.1.1.5).
(i) Trailer towing package, if required (see 3.1.1.6).
(j) Pintle height of 20 inches (+5, -0), if required (see 3.1.1.6.1).
(k) Trailer lighting cable, if required (see 3.1.1.8)
(l) Satisfactory operation on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.1).
(m) Silicone rubber hoses, if required (see 3.4.1.7).
(n) Power plant heaters and fuel warmer, if required (see 3.4.1.8).
(o) If a battery heater is not required with power plant heaters (see 3.4.1.8).
(p) A fuel fired engine preheater, if required for diesel engine driven vehicle (see 3.4.1.9).
(q) Alternator capacity, if other than as specified (see 3.4.2.2).
(r) Fuel and water separator, if required (see 3.4.3.3).
(s) Spark arrester, if required (see 3.4.4.1).
(t) Automatic transmission, if required (see 3.4.5.3).
(u) Oil lubricated wheel bearings and axle spindles, if required (see 3.4.9).
(v) Traction control, if required (see 3.4.9.1).
(w) Wide base type tires and wheels in lieu of duals, if required (see 3.4.10).
(x) Disc type wheels, if required (see 3.4.10).
(y) Bias ply, low profile or other type tires, if required (see 3. 4.10 .1).
(z) Spare tire carrier, if required (see 3.4.10.3).
(aa) Spare wheel or rim, if required (see 3.4.10.4).
(ab) Spare tire assembly, if required (see 3.4.10.5).
(ac) Trailer brake control system, if required (see 3.4.11.3).
(ad) Brake controls for use from a towing vehicle, if required (see 3.4.11.4).
(ae) Tilting hood and fender assembly or butterfly hood and bolt on fenders, if required (see 3.4.12).
(af) An individual adjustable diver’s seat, if required (see 3.4.12.1).
(ag) An air-ride driver's seat, if required (see 3.4.12.1).
(ah) Crew cab, if required (see 3.4.12.2).

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(ai) Tools, if required (see 3.4.16.1).
(aj) Gages, if required in lieu of indicators (see 3.4.19).
(ak) Automatic engine shutdown, if required (see 3.4.21).
(al) Air operated horn, in addition, if required (see 3.4.21).
(am) Front mounted winch, if required (see 3.4.22).
(an) Engine hour meter, if required (see 3.4.23).
(ao) Back-up alarm, if required (see 3.4.24).
(ap) AM/FM radio, if required (see 3.4.25).

* 6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number and vehicle type and class. Unless other conditions are cited in the contract computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Proceed Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher 1000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

* 6.4 Subject term (key word) listing.

Auger unit winch
Auxiliary engine
Boring assembly
Leveling mechanism
Power transmission
Turntable.

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6.5 Identicication of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was alone as a convenience only and the Government assumes no liability whatsoever for any inacccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

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